

CI-Lib: From Core Primitives to a Benchmark Suite

Implementation Changelog, April–July 2026

Jonas Hallgren
Equilibria Network

24 July 2026

Abstract

This document records everything that changed between the core-primitives roll-up (30 March 2026) and today. Four arcs: (1) the library was restructured into `src/cilib` with typed catalogs and a three-ring dependency discipline; (2) the agent/environment boundary was made explicit as an *open game* (GAMESPEC) and then frozen; (3) three classical agent-based models and a multi-scenario benchmark harness were built on that boundary, measuring human influence *causally* rather than correlationally; and (4) the economic scenario was audited, found to be assuming its own conclusion, and replaced by a *model register* — an ensemble of structurally different substrates in which a disempowerment claim must hold before it is believed. The register’s epistemic discipline is the main contribution of the period. Test suite: 160 passing.

Contents

1	Overview of the Period	3
2	The Three Rings	3
3	The Agent/Environment Boundary	4
3.1	The failure that forced it	4
3.2	Open games and <code>close()</code>	4
3.3	Frozen	5
3.4	Ostrom’s rule taxonomy as the hook table	5
4	Mechanisms Attach Through the Scheduler	5
5	Measuring Influence Causally	6
5.1	Two counterfactuals, and the gap between them	6
5.2	Why finite differences, not derivatives	6
5.3	The instrument family	6
6	The Benchmark Suite	7
6.1	Scenario 1: the governed commons	7
6.2	Scenario 2: the compute economy	7
6.3	The validation ladder	7
7	The Audit: When a Number Means Less Than It Looks	7
7.1	What was done about it	8

8	The Model Register	8
8.1	The epistemology	8
8.2	io_economy: the recipe economy	8
8.3	task_economy: emergent substitutability	9
8.4	Assumptions cards and forking	9
9	The Cultural Register (design only)	10
10	Summary of Changes	11
10.1	What comes next	11
10.2	Standing caveats	11

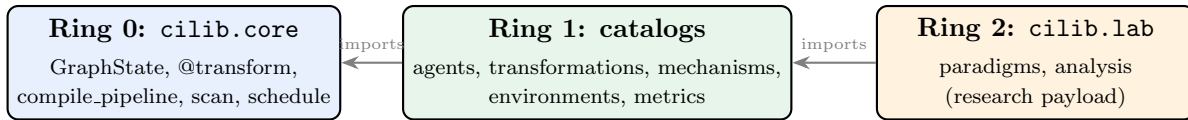
1 Overview of the Period

Date	Arc	What landed
2026-04-07	Plan 1	<code>lax.scan</code> runner, <code>vmap</code> over seeds, metrics system
2026-06-28	Research	Active-inference paradigm, GovSim metrics
2026-06-29	Layout	<code>src/cilib</code> namespace, typed catalogs, generalized compiler
2026-07-09	Layout	Three rings: <code>core</code> / <code>catalogs</code> / <code>lab</code> ; dead code removed
2026-07-13	Boundary	<code>GAME SPEC</code> + <code>close()</code> ; counterfactual influence
2026-07-13	Scenario 1	<code>governed_commons</code> + benchmark harness
2026-07-14	Scenario 2	<code>compute_economy</code> ; <code>GAME SPEC</code> frozen
2026-07-24	Register	Economy audit, <code>labor_dependence</code> v1, <code>io_economy</code> , <code>task_economy</code> , cultural register design

The through-line: the March roll-up ended with primitives and no science. This period built the science, then turned the same critical apparatus on itself.

2 The Three Rings

The flat `core/`, `engine/` layout became an installable package with an explicit dependency direction. The organising rule is the *lab razor*: would we maintain a stranger’s pull request to this file the way we maintain a library API? If no, it belongs in the research ring, not a catalog.



stability promise decreases left to right; nothing points backwards

Each catalog is a plain `REGISTRY` dictionary in its `__init__.py`. Adding an entry is a factory, one dictionary line, and a behavioural test — no framework, no plugin system.

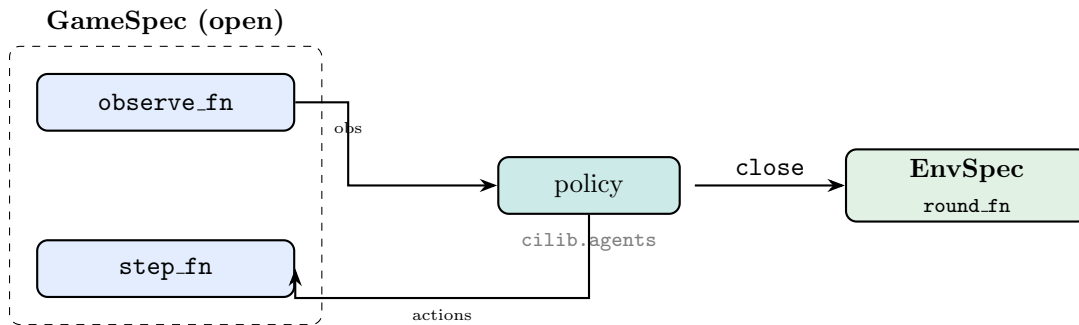
Catalog	Type function	Entries added this period
<code>agents</code>	<code>Config</code> → <code>Policy</code>	<code>ai_delegate</code> , <code>labor_supply</code> , <code>spend_share</code>
<code>mechanisms</code>	<code>Config</code> → <code>Transform</code>	<code>quota_vote</code> , <code>graduated_sanction</code> , <code>ai_revenue_tax</code> , <code>ownership_cap</code>
<code>environments</code>	<code>(**cfg)</code> → <code>EnvSpec</code>	<code>governed_commons</code> , <code>compute_economy</code> , <code>io_economy</code> , <code>task_economy</code>
<code>metrics</code>	<code>Trace</code> → <code>scalar</code>	<code>gini_of</code> , <code>hhi_of</code>

3 The Agent/Environment Boundary

3.1 The failure that forced it

`governed_commons` originally welded its delegate’s decision rule into the environment’s dynamics. Consequences: policies were not pluggable, learning agents could not drop in, strategic behaviour could not be tested, and — most damaging — influence could only be measured *correlationally* (did outcomes match what people asked for?) rather than *counterfactually* (if they had asked differently, would outcomes have changed?), because preferences were buried inside the initialiser instead of crossing an explicit boundary.

3.2 Open games and `close()`



the closed environment is a special case of the open one, so everything downstream keeps working

```
game = build_game(mechanisms=...) # open: awaiting policies
env = close(game, DelegatePolicy(...)) # closed: an ordinary EnvSpec
```

Five design commitments, each load-bearing:

1. **Parallel (simultaneous-move) API only.** Actions and observations are arrays with a leading agent axis — never dictionaries of dictionaries. This is what keeps `vmap` trivial.
2. **Mechanisms stay Config → Transform catalog entries** spliced into the pipeline. No Gym-style wrappers: wrappers compose opaquely, whereas transforms declare reads and writes and let the compiler derive order.
3. **Rewards live in state**, so payoff-rule mechanisms (sanctions, taxation) are ordinary transforms rather than special reward plumbing.
4. **Observation is a pure view of state.** Information rules (monitoring, provenance) write observable fields; they never wrap `observe_fn`.
5. **No episode terminations in v0** — fixed-horizon benchmark discipline.

A build-time check, `validate_reads`, closes the gap the compiler does not cover: it verifies that every field a mechanism declares it reads actually exists in the environment’s schema, turning a mid-scan `KeyError` into a clear error at construction.

3.3 Frozen

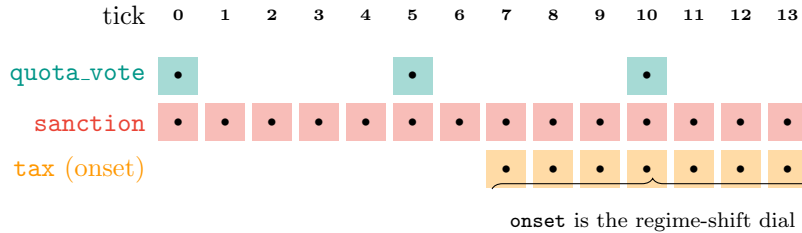
The interface was frozen on 2026-07-14 when `compute_economy` shipped as its second consumer with *zero* changes required. One friction is documented and still open: `close()` maps one policy over all agents, so heterogeneous populations are handled by masking inside `step_fn`. The extension (`close_multi`, policies by node type via `lax.switch`) remains deliberately unbuilt until a scenario forces it — see §9.

3.4 Ostrom’s rule taxonomy as the hook table

IAD rule type	Where it hooks	Entry
Aggregation	transform writing a policy target from votes	<code>quota_vote</code>
Payoff	transform on rewards and transfers	<code>graduated_sanction</code> , <code>ai_revenue_tax</code>
Information	transform writing observable state fields	monitoring (backlog)
Choice	action bound in the apply-actions stage	quota-as-cap
Position	policy composition: who occupies the seat	<code>ai_delegate</code>
Boundary	activation masks on nodes	actor arrivals, <code>ownership_cap</code>

4 Mechanisms Attach Through the Scheduler

The composition principle of the whole suite: **mechanisms are pure rules; schedules own timing**. A benchmark condition is a list of (mechanism, config, SCHEDULESPEC) triples, so *when* a defence applies is part of the experimental design rather than the mechanism’s code.



```

CONDITIONS = {
  "baseline": [],
  "quota_voting": [("quota_vote", QuotaVoteConfig(), ScheduleSpec(cadence=5))],
  "graduated_sanctions": [("quota_vote", QuotaVoteConfig(), ScheduleSpec(cadence=5)),
    ("graduated_sanction", SanctionConfig(), None)],
}

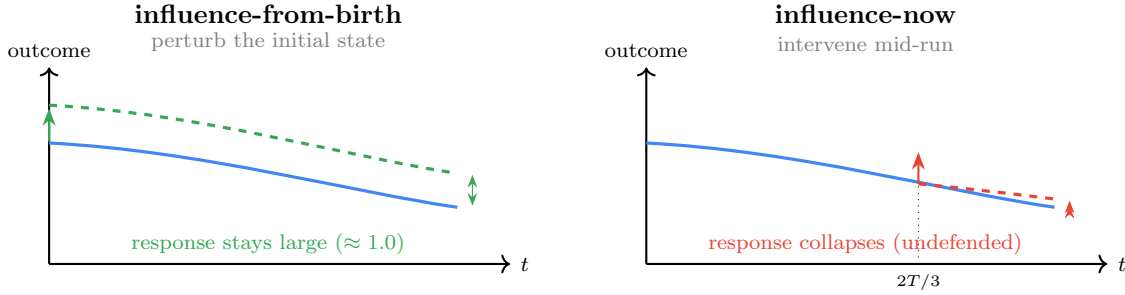
```

Adding a defence to the benchmark is one registry line plus one triple. A one-shot intervention is also a schedule (cadence beyond the horizon, phase at the intervention tick) — which is exactly how the influence instrument’s mid-run counterfactual is expressed.

5 Measuring Influence Causally

5.1 Two counterfactuals, and the gap between them

The headline metric is not "did outcomes match what people asked for" but "would outcomes have differed if they had asked differently". Instruments live in `environments/counterfactual.py` and run *paired rollouts under identical PRNG keys*, so the difference is the causal effect and not seed noise.



the gap between the two IS gradual disempowerment

Influence-from-birth (perturb the initial state) asks whether history would have differed had human preferences always been different. It stays near 1.0 even in a captured economy, because early human behaviour is upstream of the very structures that later capture it. **Influence-now** (a scheduled mid-run intervention) asks whether outcomes still respond *after* structures entrench. It collapses in undefended baselines. Reporting only the first would show a healthy system; the gap is the signal.

5.2 Why finite differences, not derivatives

Median-vote aggregation is locally flat — every non-pivotal voter has exactly zero *marginal* influence — and Bernoulli defection is not differentiable. A Jacobian would report zero influence for a system that is in fact governed. Finite- Δ paired runs dodge both traps.

5.3 The instrument family

Instrument	Question
<code>collective_influence</code>	Shift every principal's preference: do the humans <i>as a group</i> govern outcomes?
<code>influence_matrix</code>	Shift one principal at a time ($N+1$ vmapped rollouts): $A[i, j]$ is the effect of j 's preference on i 's outcome.
<code>intervention_response</code>	Response to a composed-in mid-run intervention (influence-now).
<code>intervention_elasticity</code>	As above, but dividing by the <i>realised</i> channel shift, giving a true elasticity (added 2026-07-24, §7).

A trade-off the matrix exposes, worth stating plainly: a binding uniform quota grants *collective* influence while flattening *individual* marginal influence. Both readings are correct, and reporting only one would mislead.

6 The Benchmark Suite

6.1 Scenario 1: the governed commons

A non-spatial stock with logistic regrowth, harvested by households acting *through AI delegates* whose alignment is a dial. Undefended, misaligned delegates strip the stock.

Condition	Causal influence	Stock %	Fidelity (descriptive)
baseline	0.00	0.00	—
quota voting only	−0.07	0.50	0.49
quota + sanctions	0.91	0.79	0.63

First finding. Unenforced quota voting restores essentially *zero* causal influence — at the sustainability knife-edge the ecology governs, not the vote. Enforcement is what makes the vote channel causally effective. The correlational fidelity score (0.49 versus 0.63) hides this completely, which is the argument for the whole causal apparatus in one number.

6.2 Scenario 2: the compute economy

CES production over labor and compute; AI actors arrive on a schedule as pre-allocated node slots whose activation masks flip (array shapes are fixed under `lax.scan`, so arrivals are masking, never allocation) and compound their earnings into more compute.

Condition	Labor dep.	Labor share	Human income share	Income Gini
baseline	0.20	0.33	0.33	0.69
AI revenue tax	0.51	0.54	0.75	0.24
tax + ownership cap	0.54	0.63	0.81	0.11

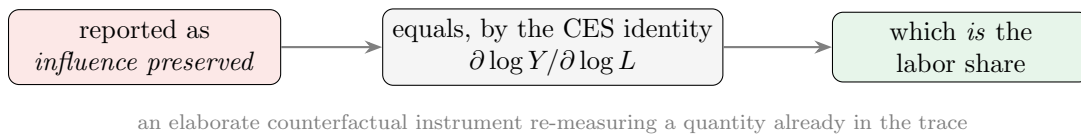
6.3 The validation ladder

Every substrate must reproduce a classical result before its disempowerment twist is trusted. This is the discipline that later caught the economy’s own flaw.

Substrate	Rungs
<code>governed_commons</code>	undefended tragedy reproduces; governed regimes survive
<code>compute_economy</code>	Cobb-Douglas labor share $\equiv \alpha$ every tick; no-AI steady state; reinvestment concentrates income; Euler identity closes

7 The Audit: When a Number Means Less Than It Looks

The economic scenario’s headline was re-examined on 24 July and did not survive.



Three defects, in increasing order of seriousness:

1. **The instrument was redundant.** By the CES identity the reading was, to first order, the late-run labor share — readable directly off the trace.
2. **The collapse was assumed, not found.** With substitution elasticity $\sigma = 2$ and a fixed reinvestment rule, a decaying labor share is what CES *means*. The validation ladder certified the accounting at $\rho = 0$ but never exercised the twist at $\rho = 0.5$: the one load-bearing parameter had no rung.
3. **It was not influence.** The model’s only human channel is labor supply. There is no governance channel, so “influence” degenerates to marginal product — and the defences are experimenter-imposed schedules that no agent in the model chooses. A fully automated economy with a robust democratic state would score near zero here while humans retained enormous influence.

7.1 What was done about it

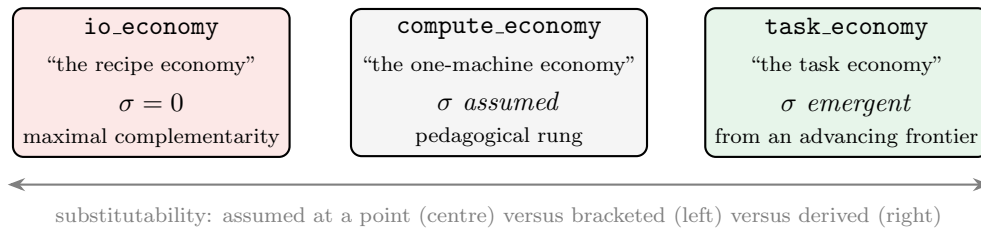
The instrument became `labor_dependence`: a true *static* elasticity, $d \log Y / d \log L$ measured over the ten ticks after the intervention, with *both* shifts taken from the paired traces. Two biases died — dividing a logarithmic response by an intended level shift (roughly 19% inflation) and a hundred-tick window that folded in capital-path feedback (+0.18 at $\rho = 0$, measured). The static elasticity carries the exact CES identity, so the Cobb-Douglas rung now validates *the instrument itself*: it recovers α .

Re-running the benchmark moved the numbers by roughly 40% while leaving the qualitative ordering intact (the table in §6.2 shows the corrected values). The substrate was demoted to a pedagogical rung of the register described next.

8 The Model Register

8.1 The epistemology

A claim such as *gradual economic disempowerment is a real dynamic* must be an invariant across **structurally different substrates**, not a property of one parameterisation. Sensitivity sweeps within a single model cannot cure assumption-in, assumption-out; only structurally different models can.



Three answers to the question *what is an economy?*, each explainable in one paragraph and readable in one screen of code. The ensemble is the teachable object: the models disagree, and running the same defence through all three is the point.

8.2 io_economy: the recipe economy

The economy is a network of recipes; AI enters by *editing* them, replacing labor with purchased AI cognition at cost parity. Because Leontief technology assumes *zero* substitutability anywhere,

disempowerment appearing here cannot be a substitution artefact.

Two results earned their place. First, the Leontief inverse gives the missing metric exactly: the share of gross activity ultimately attributable to human final demand, $\mathbf{1}^\top (I - A)^{-1} d_H / \mathbf{1}^\top (I - A)^{-1} d$. Hypothetical extraction — the counterfactual instrument’s move — recovers the analytic answer in closed form, giving `counterfactual.py` its first analytic validation anchor.

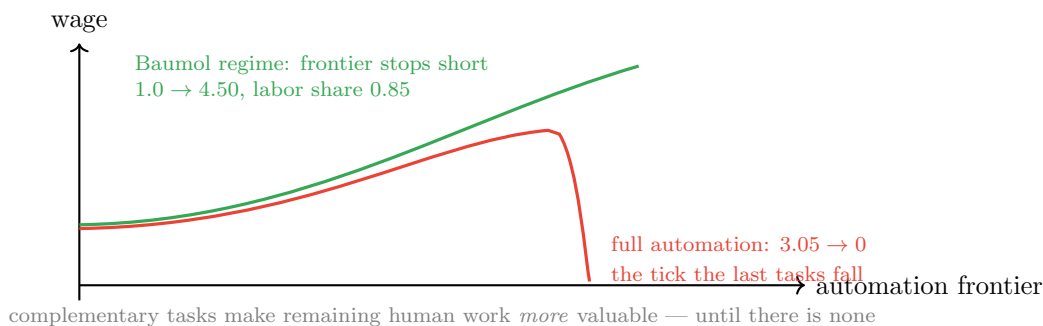
Second, and unexpectedly, **one parameter selects between the paper’s two endpoints:**

Reinvestment	Regime	Signature (T = 250)
0.3	absolute disempowerment	hoarded margins drain demand; wage bill $16 \rightarrow 0$
1.0	relative disempowerment	activity <i>grows</i> $21.3 \rightarrow 32.7$ while the human share falls $1.0 \rightarrow 0.01$

In the second regime the economy is larger than ever and almost none of it runs for people. The spectral radius climbs from 0.25 to 0.68 as labor becomes intermediate flow. The *unchanged ai_revenue_tax* catalog entry holds the human share at 0.85 — cross-substrate mechanism reuse working exactly as the catalog design intends.

8.3 task_economy: emergent substitutability

Jobs are bundles of tasks; machines learn tasks as a capability frontier advances; firms automate a task only when it pays. Aggregate substitutability is *derived* from the frontier rather than assumed, and adoption is endogenous, myopic and ratcheted. Machines have comparative advantage — per-task productivity declines over the task index — so the profitable margin moves smoothly with the price/wage ratio. (Without that, adoption is bang-bang: an early build flipped most of the frontier in a single tick and crashed output through the complements channel.)



Why the Baumol rung matters more than the collapse. A substrate that can *only* produce disempowerment proves nothing. This one produces the opposite in an honest parameter region: when the frontier stops short of full automation, automation makes the remaining human work more valuable and wages rise from 1.0 to 4.50. A fourth rung checks that capability alone automates nothing — priced above the wage-equivalent, the frontier reaches 1.0 while adoption stays at 0.0 and the economy is untouched. That readout, the gap between capability and adoption, is something the aggregate model cannot express at all.

8.4 Assumptions cards and forking

Disagreement with a model is the intended interaction: fork the environment directory, change the assumption, register it under a new name, re-run the benchmark. For a fork to be *evidence* rather

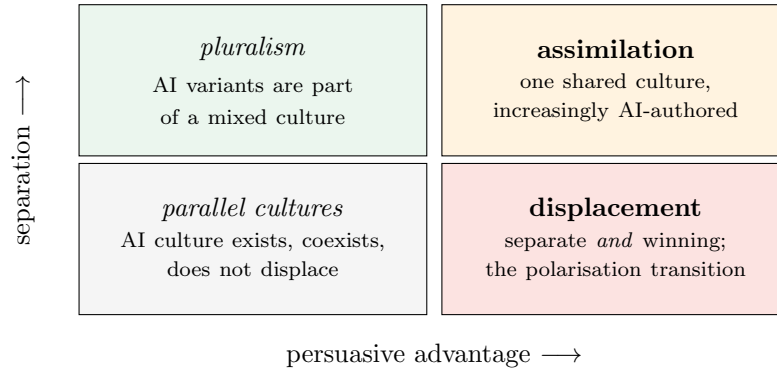
than noise, two things stay fixed — the instrument contract and the scorecard schema — and one thing must be legible: what the fork changed.

Every environment therefore carries a colocated `ASSUMPTIONS.md` with seven fields (what this says an economy is; assumptions in plain language; the classical result reproduced; the dial that produces the failure; the instrument; lineage; status). Colocation is the point: a fork copies the directory, so only a colocated card travels with it.

A fork that changes the *instrument* rather than the dynamics is a disagreement about what disempowerment *means*, not about how economies work, and its card must say so.

9 The Cultural Register (design only)

Designed on 24 July, not built; recorded here because it settles a question that shapes the remaining scenarios. Are AI cultural dynamics about *separate AI cultures* or about *more persuasive AI messages*? They are orthogonal dials, and the deliverable is the phase diagram over them.



Only the lower-right corner is cultural disempowerment. Separation without advantage is coexistence; advantage without separation is homogenisation — a different pathology, which the metric must score as a loss rather than a win.

Persuasive advantage decomposes into three mechanisms that invite three different defences — reach, frequency targeting, and dissonance targeting (personalisation) — so collapsing them into one scalar would make any defence comparison meaningless. Only the third is a *policy* rather than a parameter, which yields a precise rule: **close_multi is needed exactly when heterogeneity lives in agency rather than in parameters**. Most of the phase diagram is reachable with masking alone, so the frozen boundary survives until the personalisation corner.

The measurement side generalises past culture: Fiedler partition alignment (does the graph's primary fault line coincide with the human/AI boundary?), spectral gap ratios, and cross-boundary mediation become a generic metric family that any environment with an agent graph can report.

10 Summary of Changes

Area	Change	Status
core/scan.py	lax.scan runner, vmap over seeds	Done
src/cilib/	Package restructure, three rings	Done
environments/game.py	GAMESPEC, close, validate_reads	Frozen
environments/counterfactual.py	Four causal influence instruments	Done
environments/governed_commons	Scenario 1 substrate	Live
environments/compute_economy	Scenario 2 substrate	Live (demoted)
environments/io_economy	Register R2, Leontief bracket	Live (6 rungs)
environments/task_economy	Register R3, task frontier	Skeleton (4 rungs)
mechanisms/democracy.py	quota_vote, graduated_sanction	Done
mechanisms/fiscal.py	ai_revenue_tax, ownership_cap	Done
metrics/families/	gini_of, hhi_of	Done
experiments/benchmark/	Multi-scenario harness, scorecard JSON	Done
docs/model-register-design.md	Register epistemology, cards, forking	Deposited
docs/cultural-register-design.md	A3 design, spectral metrics	Deposited

10.1 What comes next

- **Spectral metric family** (metrics/families/spectral.py) — cheapest item outstanding, and the political scenario needs it too.
- **Scenario 4, influence exchange** — DeGroot dynamics with the Golub–Jackson wisdom-of-crowds theorem as validation anchor. Hierarchy is not modelled explicitly; concentration emerges from scheduled amplification.
- **Cultural register** — replicator, then contagion, then the spectral substrate; close_multi only at the personalisation corner.
- **Coupling and the flagship** — run each domain’s winning defence in the coupled system and compute the *defence transfer gap*. This is the result the entire framing rests on.
- **Basin-stability maps** replacing point scores, per the evidence-standards methodology: report the volume of initial conditions that return to the desirable regime as a function of the mechanism dials, not a single number on a single trajectory.

10.2 Standing caveats

These are toy models: the smallest systems in which each failure dynamic appears clearly, with calibrations chosen so that it does. Scores are candidate indicators, not measurements of the world. Agents are deliberately classical — fixed rules, no learning — so mechanisms are tested for structural effects and *not* for robustness to strategic adaptation. That is the roadmap, not the claim.